STUDY NOTES — CERTIFICATION PREP

Google Foundational Models *Study Guide*

Comprehensive technical reference on Google DeepMind & Vertex AI foundational architecture: Gemini 2.5 & 3.1 (Deep Think, context caching, grounding), generative media (Imagen 3/4, Nano Banana, Veo 2/3.1, Lyria 2/3, Chirp 2/3), Multimodal Live API, Gemma open-weight models, SynthID & Frontier Safety Framework governance, and specialized scientific models.

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01 Core Definition & Architecture

Definition: Native Multimodality (Early Fusion)

Google's Gemini foundational models are built from the ground up on **native multimodality (early fusion)**. Instead of bolting separate vision or audio encoders onto a pretrained text-only backbone—which causes serial latency, stripped nuance, and cross-modal hallucinations—all data types (text, images, audio, video, code, and PDFs) are projected into a **unified, shared token space** from the start of pretraining. This stream is processed by a single **Sparse Mixture-of-Experts (MoE)** transformer backbone, enabling real-time agentic interaction and deep cross-modal reasoning.

- **Inherent information loss in late fusion:** Converting audio or video into intermediate text captions strips paralinguistic cues (tone, sarcasm, hesitation) and fine-grained spatial nuance.
- **Compounded latency:** Chaining sequential isolated encoders (ASR → Text LLM → TTS) creates serial bottlenecks that preclude sub-second real-time agentic interaction.
- **Cross-modal hallucination:** A text-dominant LLM backbone frequently overrides visual or auditory evidence at late-fusion adapter layers when modalities conflict.

Architectural Shift: PaLM 2 vs. Gemini Sparse MoE

- **PaLM 2 Baseline (Legacy):** Dense, text-first transformer where 100% of parameters fire for every token, requiring external bolted-on Vision or Universal Speech Model (USM) encoders.
- **Gemini Sparse MoE Routing:** Conditionally routes each token to a specialized subset of "expert" feed-forward networks, decoupling total parameter capacity from active per-token FLOPs and inference latency.

02 Ecosystem Tiers

`Gemini Nano`

On-device edge model engineered for strict thermal, battery, and RAM constraints (Pixel devices via Android AICore, Chrome built-in AI) → Zero-latency, offline local inference where user data never leaves the device.

`Gemini Flash-Lite (2.5 / 3.1 / 3.5)`

Ultra-cost-efficient, high-throughput tier built for massive scale (classification, extraction, routing, translation, and lightweight agentic tasks) at minimal cost per token.

`Gemini Flash (2.5 / 3 / 3.5 / 3.8)`

High-velocity multimodal workhorse trained via **knowledge distillation** (compressing larger teacher distributions into an efficient *k*-sparse student architecture) → Delivers near-Pro coding and agentic reasoning with rapid Time-to-First-Token (TTFT).

Gemini Pro (2.5 Pro / 3.1 Pro)

Versatile frontier reasoning and coding engine balancing massive parameter capacity with a 1M–2M token context window for complex multi-step tool orchestration and repository-scale analysis.

Gemini Ultra & Deep Think (4 Argon)

Maximum-capacity frontier tier running on extensive TPU clusters for complex scientific discovery, formal mathematical verification, legal/financial synthesis, and defensive cybersecurity.

03 Gemini 2.5, 3.1 & Context Mechanics

System 2 Reasoning: Deep Think, Thinking Budgets & Thinking Levels

Modern Gemini models embed controllable internal chain-of-thought reasoning directly into the inference pipeline.

- **System 2 Reasoning ("Deep Think"):** Does not immediately emit the highest-probability surface token → Instead, it explores internal reasoning trajectories, executes code to test assumptions, and prunes erroneous paths before responding.
- **Gemini 2.5 (`thinking_budget`):** Uses an integer token cap (default dynamic –1 up to 8,192 tokens; configurable up to **24,576 tokens** on 2.5 Flash/Flash-Lite and **32,768 tokens** on 2.5 Pro). Setting `thinking_budget=0` disables thinking on 2.5 Flash and 2.5 Flash-Lite for lowest latency, whereas **thinking cannot be disabled on Gemini 2.5 Pro** (minimum 128 tokens).
- **Gemini 3 / 3.1+ (`thinking_level`):** Replaces integer budgets with discrete `thinking_level` tiers (`MINIMAL`, `LOW`, `MEDIUM`, `HIGH`) inside `ThinkingConfig`. Note: `thinking_budget` is deprecated on Gemini 3+ and Combining both `thinking_level` and `thinking_budget` in one request returns `400 INVALID_ARGUMENT`.
- **Thought Summaries & Thought Signatures:** Enable `include_thoughts=True` to inspect synthesized reasoning summaries, and return opaque **Thought Signatures** across multi-turn tool-use loops to preserve internal reasoning state (mandatory when using `MINIMAL` thinking in multi-turn conversations).
- **Context Window & Output Limit:** Supports a **1,048,576 (1M) token** input window in production (scalable to **2M tokens** in specialized environments) paired with a **65,536-token** maximum output limit.

Context Caching (Implicit vs. Explicit)

- **Implicit Caching:** Enabled by default at **zero storage cost**; automatically detects repeated prompt prefixes (up to 24h lifetime) and grants input token discounts on cache hits.
- **Explicit Caching (`cachedContents`):** Developer-managed cache resource with configurable **TTL** (default 60 minutes) and a guaranteed **90% input token discount** on Gemini 2.5+ (75% on Gemini 2.0), billed for token creation plus hourly storage. (Not supported on fine-tuned models.)
- **Isolation & ZDR:** Caches are isolated per Google Cloud project number and protected by **VPC Service Controls**. Scope caches per user session in multi-tenant agents, or disable caching entirely for strict **Zero Data Retention (ZDR)**.

Vertex AI Grounding (3 Paths)

- **Grounding with Google Search:** Dynamically queries Google's live web index at inference time to anchor time-sensitive facts and return verifiable web citations—distinct from RAG because no developer-built vector store is needed.
- **Grounding with Vertex AI Search (Enterprise Data):** Connects Gemini to up to 10 Google-managed enterprise data stores (Cloud Storage PDFs/HTML/DOCX, BigQuery, intranets) with managed ETL, OCR, chunking, and IAM (`roles/discoveryengine.viewer`).
- **Vertex AI RAG Engine:** Modular DIY orchestration over custom corpora and vector databases (Vertex AI Vector Search / ScaNN, AlloyDB, Elasticsearch, Pinecone).

Context Mechanics & Multimodal Tokenization

- **Text:** 256,000-token vocabulary using **SentencePiece** unigram tokenization.
- **Visual (Images):** Dynamic tiling. Images are scaled/cropped into **768×768** pixel tiles; each tile deterministically maps to **258 tokens**.
- **Video:** Parallel streams sampled at configurable rates (**1 FPS ≈ 263 tokens/sec**, compressible to **66–70 tokens/sec** for multi-hour recordings).
- **Audio:** Encoded continuously at a highly efficient fixed rate of **32 tokens per second**.
- **Timestamp Tokens:** Interleaved timestamp markers anchor parallel video and audio streams for exact temporal coherence.

04 Generative Media Models

Imagen 3 / 4 & Nano Banana 2 (Gemini Native Image Generation)

- **Imagen 3 & Imagen 4 (`imagen-4.0-generate-001`, `Ultra`, `Fast`):** Specialized latent-diffusion text-to-image engines (480-token prompt limit, up to 2K resolution, 1–4 images per request) optimized for low-latency studio photorealism, complex lighting, brand logos, and crisp typography with built-in SynthID and C2PA Content Credentials.
- **Nano Banana 2 (`gemini-3.1-flash-image`), Nano Banana Pro (`gemini-3-pro-image`) & Nano Banana 2.1:** Native multimodal image generation and editing integrated into the Gemini backbone (the recommended migration target for legacy standalone `imagegeneration` / `imagen` endpoints).
- **Extended Context & Entity Resolution:** Features a **131,072-token** input context, ingests PDF and multi-image references, outputs up to **4K resolution**, and preserves structural identity across **up to 5 characters and 14 objects** in multi-turn mask-free conversational editing.
- **Search-Grounded Visuals:** Executes live Google Search queries during generation to render accurate text (~94% accuracy), data visualizations, and architectural landmarks → bypassing standard diffusion hallucination.

Veo 2 & Veo 3.1 (Video + Native Audio Generation)

- **Evolution from Veo 2 to Veo 3.1:** Evolved from **Veo 2** (silent high-resolution video generation, camera-motion controls, and mask-based inpainting/outpainting) to **Veo 3 / Veo 3.1** with native audio-visual synthesis.
- **Joint Latent Diffusion:** Simultaneously compresses raw video pixels and raw audio waveforms into a single unified latent representation → Emergent, frame-accurate synchronization (footsteps, dialogue lip-sync, and ambient Foley align with visual impact without a bolted-on audio track).
- **Temporal Permanence & Formats:** Uses spatial and temporal attention for strict object permanence across occlusions ("AI slop" prevention). Supports 4, 6, or 8-second durations, up to **4K resolution**, and native **16:9 landscape** and **9:16 portrait** aspect ratios.
- **Model Variants: Veo 3.1 Generate** (maximum visual fidelity for final cuts), **Veo 3.1 Fast** (low-latency production workflows), and **Veo 3.1 Lite** (cost-effective, high-volume developer API tier).

Lyria 2 & Lyria 3 (Audio & Music Generation)

- **Temporal Latent Diffusion:** Abandons autoregressive token prediction (which causes long-horizon statistical drift) → An autoencoder compresses 16-bit PCM waveforms into compact latents, and a transformer-based denoising network iteratively sculpts pure noise into high-fidelity stereo waveforms (44.1kHz–48kHz).
- **Tiers: Lyria 2 (`lyria-002`)** (GA text-to-music), **Lyria 3 Clip (`lyria-3-clip-preview`)** (30-second rapid prototyping), and **Lyria 3 Pro (`lyria-3-pro-preview`)** (184-second full compositions).
- **Multimodal Input & Structural Control:** Accepts both **text and image inputs** (performing aesthetic extraction to generate a soundscape matching the visual mood) plus explicit mathematical structure controls (tempos like `120 BPM`, intros/choruses, intensity arcs) and multilingual expressive vocals with time-aligned lyrics.

05 Gemini Audio, Chirp 2/3, Live API & Embe

Gemini Audio (Native Speech-to-Speech)

- **Native Speech-to-Speech Architecture:** Bypasses cascaded ASR → LLM → TTS pipelines → Directly ingests and outputs audio tokens, preserving tone, emotion, pacing, and intent lost in text transcription.
- **Spoken Function Calling:** Achieves **71.5% on ComplexFuncBench Audio**, reliably triggering multi-step external tools directly from spoken audio without transcription pauses.
- **Live Speech Translation & TTS:** Streams speech-to-speech translation with native prosody/style transfer across languages, complemented by studio-grade **Gemini 3.8 Flash TTS** and **Flash-Lite TTS**.
- **Instruction Adherence: 90% strict instruction adherence rate** with built-in acoustic noise filtering for multi-turn consistency in loud or outdoor environments.

Chirp 2 & Chirp 3 (USM Speech-to-Text V2)

- **Universal Speech Model (USM) Backbone:** Built on Google's **2B+ parameter USM** foundation trained via self-supervised learning (BEST-RQ) across 12M+ hours of speech and 100+ languages.
- **Cloud Speech-to-Text V2 (`chirp_3`):** Supports streaming (`StreamingRecognize`), synchronous (`Recognize`), and long-form batch (`BatchRecognize`) transcription with speaker diarization and word-level timestamps.
- **Enterprise Controls:** Features **Custom Prompt formatting** (natural-language date/capitalization rules), **Model Adaptation** (up to 1,000 domain phrases), a built-in **Audio Denoiser** (`denoise_audio=True` for rain/traffic/music while preserving human speech), and tunable `endpointing_sensitivity`.

Gemini Multimodal Live API

- **Stateful Bidirectional WebSockets (WSS):** Replaces stateless HTTP REST calls with persistent, low-latency WebSocket connections (`gemini-2.5-flash-live-api`, `gemini-3.8-live`).
- **Simultaneous Streaming:** Ingests raw **16-bit PCM audio**, video frames, and text concurrently while outputting raw **24kHz audio** directly.
- **Barge-In & Affective Dialog:** Server-side Voice Activity Detection (VAD) lets users interrupt the model mid-sentence (**barge-in**) while mirroring emotional tone.

Gemini Embedding & MRL

- **Matryoshka Representation Learning (MRL):** Both `gemini-embedding-001` (text/code up to 3,072 dims) and `gemini-embedding-2` allow truncating `output_dimensionality` (e.g., to **1,536, 768, 512, 256, or 128**) to shrink vector index RAM/storage costs with minimal recall loss.
- **Unified Multimodal Vector Space (`gemini-embedding-2`):** Maps text, images, audio, video (with `fps` & `audio_track_extraction`), and PDFs (`document_ocr`) into a single L2-normalized vector store for cross-modal retrieval.

06 Gemma Open-Weight & Edge Models

Gemma 2, Gemma 3 & Gemma 3n / 4 Architecture

- **Gemma 2 (2B, 9B, 27B):** Text-only open-weight models introducing alternating local/global attention, Grouped-Query Attention (GQA), Logit Soft-Capping, and knowledge distillation from larger teacher models.
- **Gemma 3 (1B, 4B, 12B, 27B) Native Multimodality:** Uses a frozen **400M-parameter SigLIP Vision Transformer** on 4B/12B/27B variants to condense each image crop into **256 "soft tokens"** prepended to text sequences across 140+ languages.
- **Pan & Scan Adaptive Windowing:** Dynamically segments high-resolution or non-square images into **896×896** crops during inference without forcing aspect-ratio distortion.
- **128K Context & 5:1 Interleaved Attention:** Replaces uniform global attention with **5 layers of local sliding-window self-attention (1,024 tokens) per 1 layer of global self-attention** → Geometrically shrinks the KV-cache memory footprint.
- **Gemma 3n (E2B, E4B) & Gemma 4 (E2B, E4B, 26B A4B MoE, 31B):** Mobile-first Per-Layer Embedding (PLE) / MatFormer architectures supporting **text, image, video, and native audio** inputs on low-RAM edge devices.

RecurrentGemma (Griffin Architecture)

- **Hybrid Gated Linear Recurrence:** Replaces quadratic attention with **Real-Gated Linear Recurrence (RG-LRU)** + local sliding-window attention.
- **Fixed-Size Recurrent State:** Eliminates unbounded KV caches on long sequences.
- **Trade-off:** Slight drop in exact needle-in-a-haystack retrieval → Substantially higher sampling throughput with constant memory at the edge.

Specialized Gemma Family

- **ShieldGemma & ShieldGemma 2 (2B, 4B, 9B, 27B):** Open safety classifiers scoring text prompts/responses and images (ShieldGemma 2 4B) against harm policies.
- **PaliGemma & PaliGemma 2 (3B, 10B, 28B):** Open VLMs (SigLIP + Gemma) for fine-tuning on VQA, captioning, OCR, object detection, and segmentation.
- **CodeGemma (2B, 7B):** Open Fill-in-the-Middle (FIM) code completion and coding instruction models.
- **MedGemma (4B, 27B) & TxGemma (2B, 9B, 27B):** Open medical text/image comprehension and therapeutic drug-discovery models.

07 Safety, SynthID & Frontier Governance

SynthID Imperceptible Watermarking Framework

Traditional metadata watermarks (EXIF or C2PA manifests) are easily stripped during re-encoding or screenshots. **SynthID** embeds imperceptible cryptographic signatures directly into the generated payload at inference time (paired with C2PA Content Credentials):

- **Text (Tournament Sampling):** Uses a cryptographic pseudo-random ***g*-function** seeded per token to run a multi-layer tournament that subtly skews logit token probabilities without degrading fluency or perplexity.
- **Audio (Psychoacoustic Spectral Embedding):** Converts waveforms to spectrograms and embeds watermark bits in frequency bands masked by human hearing → Survives MP3 compression and the **"analog hole"** (playing through a physical speaker and re-recording with a microphone).
- **Video & Image (Spatial-Temporal Latent Manipulation):** Distributes watermark signals holistically across pixel latents and spatio-temporal volumes → Survives cropping, color-filtering, overlays, and lossy compression.

Frontier Safety Framework (FSF) & Vertex AI Guardrails

- **Google DeepMind Frontier Safety Framework (FSF):** Capability-gated governance protocol defining empirical **Critical Capability Levels (CCLs)** across four catastrophic risk domains: **CBRN uplift**, **Autonomous Cyber-Offense**, **Machine Learning R&D / Autonomous Self-Improvement**, and **Deceptive Alignment / Circumvention of Human Oversight**.
- **Coupled FSF Mitigations:** Crossing early-warning thresholds triggers mandatory **Security Mitigations** (hardened weight storage, egress throttling, and multi-party authorization against exfiltration) and **Deployment Mitigations** (automated sentinel classifiers, safety fine-tuning, and staged access gates).
- **Vertex AI Safety Filters & Model Armor:** Per-request `SafetySetting` harm-block thresholds across Hate Speech, Harassment, Sexually Explicit, and Dangerous Content, reinforced by **Vertex AI Model Armor** and **ShieldGemma 2** for enterprise prompt-injection, jailbreak, and PII/DLP screening.

08 Scientific & Domain-Specialized Models

AlphaFold 3 & AlphaProteo

- **AlphaFold 3 Paradigm Shift:** Replaces the MSA-heavy Evoformer with the **Pairformer** (focusing directly on pairwise atomic interactions) and tokenizes chemical modifications and ligands at **one token per individual atom**.
- **Conditional Diffusion Module:** Replaces AF2's rigid deterministic structure module → Denoises a chaotic 3D cloud of random atom coordinates into joint complexes across **proteins, DNA, RNA, ligands, and ions**.
- **AlphaProteo (De Novo Binder Design):** Generative biology system that designs novel, high-affinity **protein binders** for disease target sites (e.g., VEGF-A, viral spike proteins), achieving 3×–300× higher binding affinities.

AlphaGeometry 2, MedLM & Sec-PaLM

- **AlphaGeometry 2 (Neuro-Symbolic Proving):** Solves **84% of IMO geometry problems** by pairing a heuristic **Gemini neural language model** (predicting auxiliary geometric constructs like unstated bisectors) with the infallible **DDAR symbolic engine** (exhaustive rule-bound deduction).
- **Med-PaLM 2 → MedLM → MedGemma:** Clinical foundation models (~86.5% USMLE expert pass rate) for clinical Q&A, chart summarization, and medical imaging; consolidated on Vertex AI into multimodal **Gemini 2.5/3** and open-weight **MedGemma (4B/27B)** + **MedSigLIP**.
- **Sec-PaLM 2 → Gemini Cyber:** Security models trained on Mandiant & Google Threat Intelligence telemetry for malware script analysis and SIEM operations, succeeded by **Gemini 3.8 Flash Cyber** and **Gemini 4 Argon Cyber**.

Scientific, Robotics & Specialized Model Reference

`SIMA 2 (generalist 3D agent)`

Collaborative generalist agent that perceives screen pixels, reasons, and follows natural-language instructions across diverse virtual 3D worlds.

`Genie 2 / 3 (world models)`

Generates interactive, playable, physics-consistent 2D and 3D environments on the fly from a single image or textual prompt.

`AlphaFold 3 (biology)`

Predicts joint 3D molecular structures and interactions across proteins, DNA, RNA, ligands, and ions using Pairformer + diffusion.

`AlphaProteo (biology)`

Designs novel, high-affinity protein binders from scratch for diverse target proteins to accelerate drug discovery.

`AlphaMissense (genomics)`

Classifies 71M human missense mutations to identify pathogenic genetic variants using structural context.

DolphinGemma (bioacoustics)

Specialized acoustic sequence model for decoding and modeling wild marine mammal communication patterns.

AlphaGeometry 2 (mathematics)

Solves 84% of IMO-level geometry problems using a neuro-symbolic dual-engine approach (Gemini + DDAR).

AlphaTensor (mathematics)

Uses deep RL to discover provably exact, computationally faster algorithms for fundamental matrix multiplication.

AlphaCode / AlphaCode 2

Generates competitive-programming solutions through massive-scale Gemini sampling, execution filtering, and clustering.

AlphaDev (algorithms)

Utilizes reinforcement learning at the CPU assembly level to discover faster sorting and hashing algorithms in C++ LLVM.

AlphaGo / AlphaZero / MuZero

Frontier RL models progressing from human games to zero-data self-play (AlphaZero) to planning without known rules (MuZero).

AlphaStar (games)

First AI to defeat Grandmaster-level players in the imperfect-information real-time strategy game StarCraft II.

GraphCast / GenCast (climate)

Delivers 10-day deterministic (GraphCast GNN) and 15-day probabilistic (GenCast diffusion) global weather and extreme-event forecasts.

AlphaChip (hardware)

Reinforcement learning placement engine that optimizes superhuman TPU and ASIC chip floorplans in hours.

AlphaQubit (quantum)

Recurrent transformer neural decoder that identifies and corrects errors in surface-code quantum processors.

PaLM 2 / USM / WaveNet

Foundational dense text-first LLM (PaLM 2), 2B-parameter Universal Speech Model (USM), and raw-audio synthesis (WaveNet) baselines.

Phenaki (video)

Early causal video transformer for generating long, temporally consistent video sequences from time-varying text prompts.

Aeneas / Ithaca (humanities)

Restores missing text, attributes geographic origin, and dates damaged ancient Greek and Latin inscriptions.

PaLM-SayCan / RT-2 / Gemini Robotics

Vision-Language-Action (VLA) robotics models translating visual camera feeds and language goals directly into motor actions.